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PALEOMAGNETIC EVIDENCE FOR POST-2.55-GA TECTONIC TILTING AND 1.1-GA REACTIVATION IN THE SOUTHERN KAPUSKASING ZONE, ONTARIO, CANADA

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Abstract. We report paleomagnetic results from two traverses in the southern part of the Kapuskasing structural zone (KSZ) of north-central Ontario, which is hypothesized to be an oblique cross section through upper and middle Archean crust exposed by uplift and tilting. Anorthosites and granulite-grade gneisses were sampled to the east in the Shawmere traverse, while amphibolite-grade tonalitic gneisses were sampled to the west in the Racine traverse. Three distinct remanent magnetization (RM) components were isolated by alternating field and thermal demagnetization. The A RM has mean directions D = 50°, I = +62° (k = 138, α_{95} = 5°, N = 5 sites (23 samples)) for the Shawmere traverse and D = 23°, I = +53° (k = 49, α_{95} = 7°, N = 8 sites (46 samples)) for the Racine traverse. Because of the high (>500°C) laboratory unblocking temperatures of the A RM in the Shawmere granulite-grade rocks, we believe this magnetization was acquired in early slow uplift, without tilting, from about 24-km to about 16-km depths, and has survived the later main KSZ uplift and tilting event. The Racine A RM has laboratory unblocking temperatures down to ≈400°C and was blocked prior to uplift from a shallower crustal level (about 8 km). With tilt corrections of 30° and 15°, respectively, about the 35°E striking axis of the KSZ, the Shawmere and Racine A paleomagnetic poles fall on the Laurentian apparent polar wander path around 2.55 Ga, dating the time of initial uplift. Crustal tilting in the KSZ occurred after 2.55 Ga and was greater in the east than in the west. The B RM, found mainly in Shawmere anorthosites, has D = 115°, I = -68° (k = 17, α_{95} = 17°, N = 4 sites (14 samples)), with a paleopole at 144°W, 49°N, near 1.1 Ga Keweenawan reverse poles and the normal pole of the nearby 1.0 Ga Nemegosenda carbonatite. The B RM is probably a thermochemical overprint related to reactivation of the KSZ at the time of Keweenawan rifting and volcanism in the Lake Superior region. A novel C RM, with D = 64°, I = +2° (k = 18, α_{95} = 13°, N = 6 sites (10 samples)), is either an unresolved hybrid of the A and B RMs or an earliest Keweenawan (1.3 Ga) overprint.

Introduction

The composition and structure of the lower continental crust have been inferred mainly from remotely sensed geophysical properties and petrological and geochemical study of xenoliths in diatremes and volcanic pipes [Fountain and Salisbury, 1981]. The key to a better understanding of deep crustal lithologies and structural relationships lies in the study of uplifted zones at ancient convergent plate margins, where erosion of the upper plate has exposed a cross section of the continental crust [Wasilewski and Fountain, 1982; Williams et al., 1985]. Such zones are marked by anomalies in seismic structure, gravity, and aeromagnetics, and typically have a gradational decrease in metamorphic grade away from major fault zones along which uplift took place.

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The Kapuskasing structural zone (KSZ) in north-central Ontario is believed to represent one of these crustal cross sections (Figure 1). The KSZ is a 400-km long, NNE trending belt of dense high-grade gneissic rocks, cutting across the E-W grain of greenschist and amphibolite-grade rocks of the Archean Superior Province, from the late Proterozoic margin of Lake Superior in the south to the Hudson Bay lowlands, where it is buried under Phanerozoic cover. To explain the gravity signature and WNW gradational decrease in metamorphic grade, Percival and Card [1983] proposed that the southern part of the KSZ represents an oblique cross section through the upper two thirds of the Archean crust, uplifted along a listric fault whose surface expression is the Ivanhoe Lake cataclastic zone (ILCZ) (Figure 2). High-grade rocks of the KSZ are juxtaposed with greenschist-facies rocks of the Abitibi Subprovince across the ILCZ, whereas the western boundary of the KSZ with the Wawa Subprovince is gradational, implying premetamorphic continuity.

Our paleomagnetic sampling was carried out along two traverses in the southern KSZ between Chapleau and Foleyet (Figure 2). The Racine traverse near the KSZ-Wawa Subprovince boundary sampled amphibolite-grade tonalitic gneisses, which Percival and Card propose represent a 10 — 15 km thick middle crustal layer. The Shawmere traverse was in granulite-grade, lower-crustal rocks 20 — 25 km to the east. The northern line of this traverse crossed the main body of the Shawmere anorthosite, while the southern line sampled highly foliated tonalites and paragneisses just west of the ILCZ. Our purpose was to study the thermal and structural history of these two crustal sections of different original levels.

Geology, Structure, and Evolution of the KSZ

The Shawmere anorthosite consists of a detached southern mass (5x15 km) of coarse gabbroic anorthosite and a 15x50 km main body with various lithological and textural units. It was probably intruded into older tonalites and paragneisses at or before 2.765 Ga [Percival and Krogh, 1983], based on the U/Pb zircon age of a garnetiferous tonalite within the Shawmere main body.

A number of 1- to 2-km diameter carbonatite plugs and associated lamprophyre dikes occur along a 170-km-long belt in the KSZ. There were apparently three episodes of incipient rifting and alkaline intrusion, around 0.6, 1.1, and 1.7 Ga (Gittins et al. [1967]: whole rock K/Ar).

Structural characteristics of the southernmost (Chapleau) block of the KSZ which are of paleomagnetic importance are the prominent NNE grain and the hypothesized overall WNW dip of the various crustal layers, originating from thrusting and uplift. Dip angles have been estimated from geobarometry and from the attitudes of pre-uplift dikes. Pressures of 5.4 — 8.4 kbar, corresponding to depths of 18 — 28 km, for rocks from the western and eastern KSZ require a dip of ≈10° across the entire KSZ [Percival and Card, 1983]. However, a steeper dip of 15-20° in the eastern KSZ is implied by the easterly dip of late Archean Hearst dikes from the KSZ compared with the near-vertical to steep westerly dip of Hearst dikes NW of the KSZ and correlative Matachewan dikes in the Abitibi Subprovince

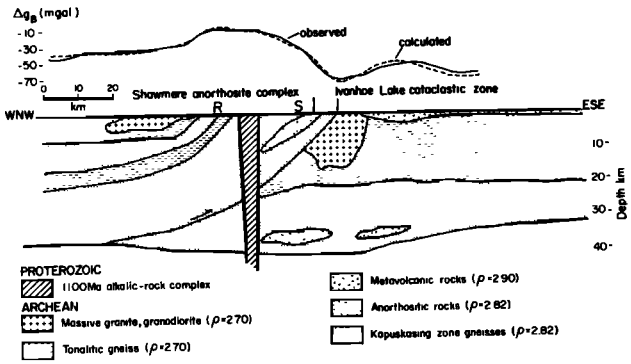


Fig. 1. Bouguer gravity profile (in mGal) and model cross-section of the southern KSZ, showing listric faulting along the Ivanhoe Lake cataclastic zone and its continuation at depth [after Percival and Card, 1983]. The dip of the thrust plane and amount of crustal tilt are thought to decrease to the west. R and S mark the approximate locations of the Racine and Shawmere traverses, respectively.

[Ernst and Halls, 1984]. A westward shallowing of the dip of crustal layers, as sketched in Figure 1, would reconcile the two estimates [Percival and Card, 1983]. The dip of the subsurface extension of the ILCZ is 35-38° WNW from preliminary seismic reflection profiles [Cook, 1985].

The late Archean Kenoran orogeny marked the end of widespread folding, intrusion, and metamorphism in the Superior Province. Following sedimentation and volcanism in the Abitibi and Michipicoten greenstone belts between 2.75 and 2.70 Ga [e.g., Nunes and Jensen, 1980], major deformation of the Wawa and Abitibi subprovinces is bracketed by the ages of the youngest volcanism (2.70 Ga) and the intrusion of late syntectonic plutons (2.68 Ga) [Percival and Sage, 1984]. However, there is evidence of continued activity in the KSZ after that time. Mafic gneiss and paragneiss samples near the ILCZ have given U/Pb zircon ages of 2.65-2.63 Ga [Percival and Krogh, 1983], but these ages probably record cooling to about 700°C and the blocking of Pb diffusion in zircons rather than a second metamorphic event.

A second period of activity from 1.85 to 1.65 Ga, perhaps related to events in the Trans-Hudson orogen to the north, has been proposed by Percival and Sage [1984] as the likely time of the main uplift. Late Proterozoic $^{40}Ar/^{39}Ar$ ages from minerals with low Ar blocking temperatures [Farrar and Archibald, 1985; Hanes, 1985; Lopez-Martinez and York, 1986] suggest a mild reactivation at 1.1-1.0 Ga.

Sampling, Measurements, and Analysis

Five or more samples oriented by Brunton compass were collected from 15 sites on the Racine and Shawmere traverses. The eight Racine sites (46 samples analyzed) were in tonalites and mafic gneisses, while five of the Shawmere sites (27 samples) were in the anorthosite, and two sites (15 samples) were in gneisses of the southern line. Stepwise alternating field (AF) demagnetization to 100 mT and thermal demagnetization to 600°C were carried out in magnetic shields with standard commercial demagnetizers. Magnetizations were measured with Schonstedt and Digico magnetometers.

The characteristic (highest coercivity H_C or laboratory unblocking temperature T_{UB}) RM component, or ChRM, of each specimen was revealed as a stable endpoint direction on stereographic projections. Minimum dispersion of directions over a demagnetization interval in which RM

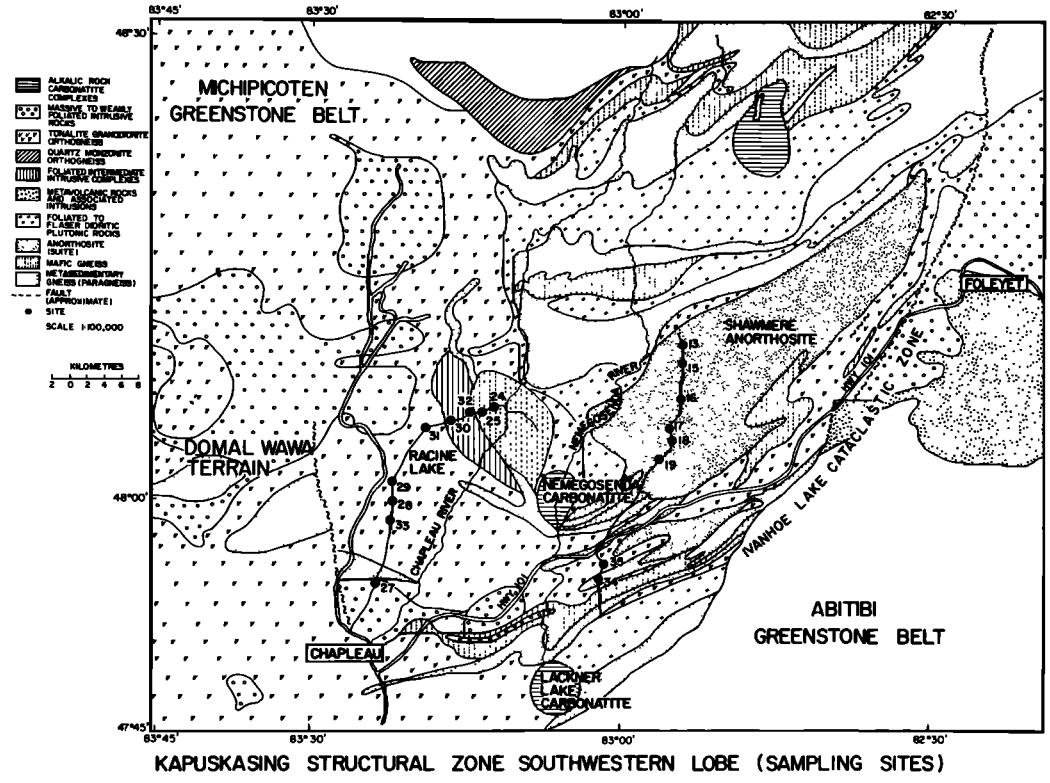


Fig. 2. Sketch map of the major geological units in the southern (Chapleau) block of the KSZ, with the sampling sites and traverses marked.

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intensity decayed significantly was required in defining the stable endpoint. Lower H_C or T_{UB} RM components yielded linear trajectories on orthogonal vector projections [Zijderveld, 1967]. Trajectories were identified by eye but were only accepted if defined by at least four successive points with an angular uncertainty [Kirschvink, 1980] of less than 7°.

Paleomagnetic and Rock Magnetic Results

Three different groups of remanent magnetizations, which we call for simplicity A, B, and C, were isolated by AF and thermal demagnetization. In some cases, two of these RMs were present in a sample, giving a multicomponent NRM, but usually one of the components dominated the NRM after removal of soft present field overprints. Thermomagnetic analyses of selected specimens showed that nearly pure magnetite with a Curie point near 580°C was the main carrier of all three RM components. However, susceptibility determinations on about 250 specimens indicate a wide range of domain states. Königsberger Q_n ratios between remanent and induced magnetization ranged from 0.03 to >50. Those anorthosites, mafic gneisses, and tonalites which proved to be paleomagnetically stable recorders tended to have the higher Q_n ratios, indicating single-domain-like behavior. The most detailed rock magnetic information about the nature and stability of RM carriers comes from coercivity and unblocking temperature spectra determined from AF and thermal demagnetization of NRM. These results will be discussed separately for each component.

Component A Results

The A component is the ChRM of most tonalites and mafic gneisses and some anorthosites from the Racine and Shawmere traverses. It has a NE declination, positive inclination direction which is distinct from both present Earth's field (PEF) and dipole field directions.

Figure 3 illustrates the typical demagnetization response

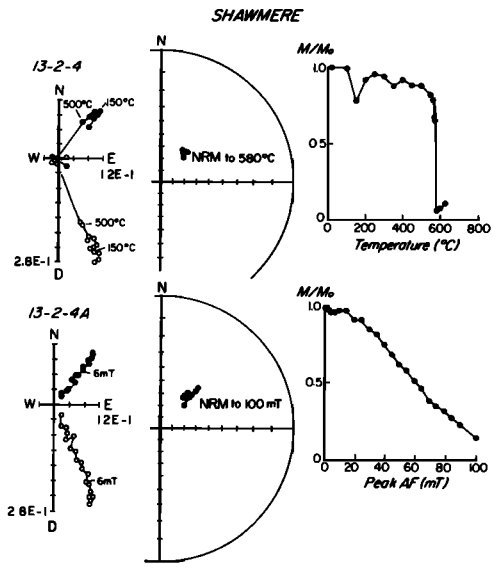


Fig. 3. Typical high unblocking temperatures and coercivities of the A RM in anorthosite samples. Orthogonal vector and stereographic projections of magnetization vectors in the course of AF and thermal demagnetization and the corresponding total intensity decay curves are shown for two specimens of a Shawmere anorthosite sample from site 13.

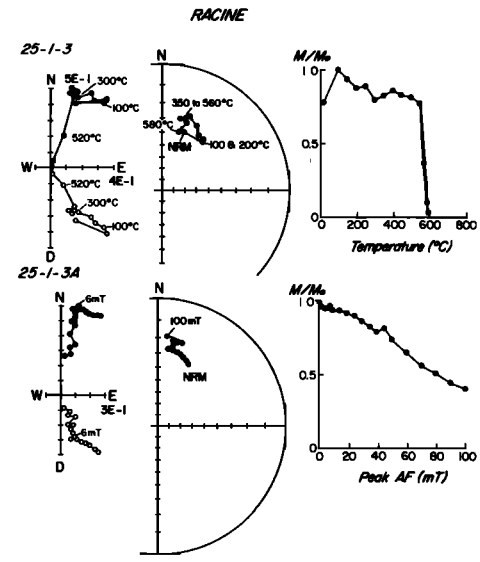


Fig. 4. AF and thermal demagnetization of a high H_C, high T_{UB} A magnetization with a low H_C, low T_{UB} PEF overprint in two specimens of a Racine mafic gneiss sample from site 25.

of the A magnetization in anorthosite samples. The RM is univectorial, with high (15 — 100 mT) coercivities and discrete T_{UB}s just below 580°C, the Curie temperature of magnetite. The AF demagnetization curve is of single-domain form, with an initial plateau, inflected shape, and high median destructive field (MDF).

Racine mafic gneiss samples exhibit a similar high coercivity, high unblocking temperature A ChRM, of single-domain type (Figure 4). On this is superimposed a low H_C, low T_{UB} component, probably a viscous remanence acquired in the PEF.

Racine tonalite samples often have a directionally fairly stable, apparently univectorial A RM (Figure 5). However, the H_C and T_{UB} spectra are bimodal. Coercivities below 20 mT and unblocking temperatures below about 400°C seem to be associated with multidomain magnetite, to judge by the initial exponential decay of

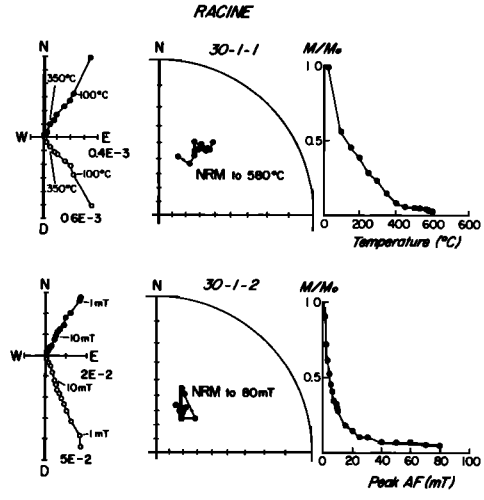


Fig. 5. AF and thermal demagnetization of an A RM exhibiting broad coercivity and unblocking temperature spectra in a Racine tonalite sample from site 30.

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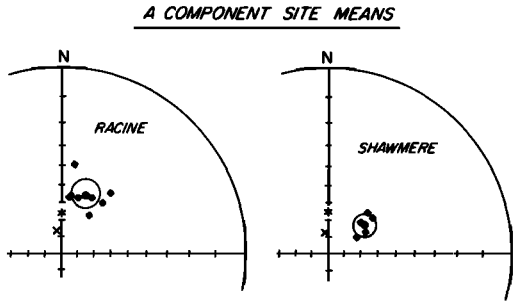


Fig. 6. Stereographic projection of site mean A directions (diamonds) for the Racine and Shawmere traverses, together with the traverse means and circles of 95% confidence. The PEF and dipole field directions are indicated by crosses and stars, respectively.

NRM in AF cleaning. The most resistant 10-20% of the NRM has a slow AF decay and unblocking temperatures concentrated just below 580°C, and probably resides in single-domain like magnetite. It is probable that only the hard fraction represents the A component isolated in the demagnetization data of Figures 3 and 4. In the thermal demagnetization vector plot, the first-demagnetized fractions have a steeper inclination and more northerly declination than more resistant fractions. The soft fraction is likely a PEF overprint, from which the harder A RM is poorly resolved because of its smaller intensity and similar direction.

Site mean directions of the A component are plotted in Figure 6 for both traverses. For the five Shawmere sites averaged, the mean A direction is D = 50°, I = +62° (k = 138, α_{95} = 5°) and for the eight Racine sites, D = 23°, I = +53° (k = 49, α_{95} = 7°) (Table 1). Vandall [1984] and Symons and Vandall [1985] reported similar RM directions from samples taken along Highway 101 between Chapleau and Foleyet.

TABLE 1. Component A Site Mean Directions After AF and Thermal Cleaning

Site	N	D, deg	I, deg	k	α_{95}, deg
		Racine Traverse			
24	4	7	57	18	16
25	6	8	39	71	7
28	3	17	56	131	7
29	5	40	52	139	5
30	16	29	53	18	8
31	5	37	62	85	8
32	2	40	45	67	
49	5	10	56	36	10
Mean	8	23	53	49	7
		Shawmere Traverse			
13	5	43	56	131	6
16	2	45	63	344	
18	1	50	56		
34	12	59	64	20	9
35	3	59	70	44	14
Mean	5	50	62	138	5

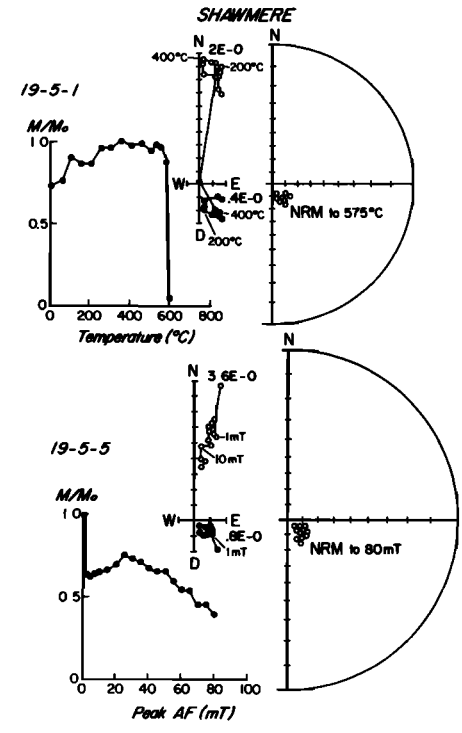


Fig. 7. AF and thermal demagnetization of the high H_C, high T_{UB} B RM of specimens of a Shawmere anorthosite sample from site 19.

Component B Results

Component B, with an easterly, steeply up direction, is the ChRM of many anorthosite samples. The B magnetization is in some cases a high H_C, high T_{UB} RM (Figure 7). In other cases, although it is directionally fairly stable, the B RM has bimodal H_C and T_{UB} spectra and seems to comprise two vectors with slightly different directions (Figure 8). In this case, both the soft multidomain fraction and the resistant single-domain like fraction carry essentially the same B remanence. PEF overprinting is minor.

Site mean directions for the B magnetization are plotted in Figure 9. Their average is D = 115°, I = -68° (k = 17, α_{95} = 17°, N = 4 sites) (Table 2). Vandall [1984], Symons and Vandall [1985], and Ernst and Halls [1984] also found easterly, steeply up RMs in tonalites, anorthosites, and some mafic dikes from the Chapleau-Foleyet area.

Component C Results

Component C, with a shallow easterly direction, occurs sporadically in anorthosites, tonalites, and mafic gneisses of both the Racine and Shawmere traverses. It has not been reported previously from the Kapuskasing area.

Two examples of the C magnetization are shown in Figure 10. In anorthosite sample 15-2, the C RM has intermediate coercivities but high laboratory unblocking temperatures. In Racine mafic gneiss sample 24-3, the C component is a relatively high H_C, high T_{UB} RM, overprinted by a soft, perhaps PEF component. In both cases, the intensity decay plots are more irregular than those of the A or B RMs.

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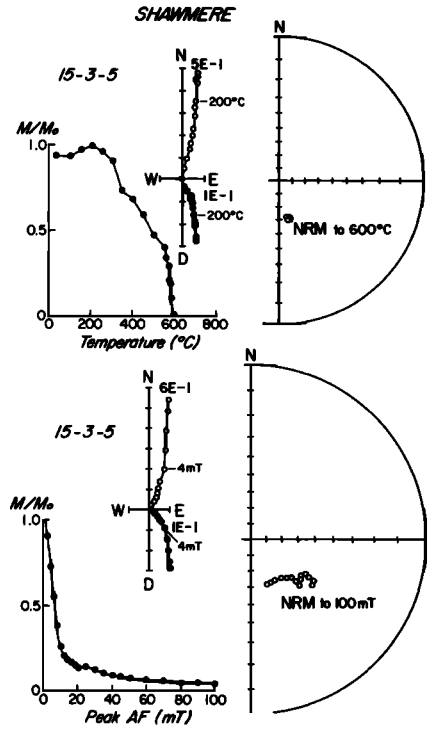


Fig. 8. AF and thermal demagnetization of a B RM with distributed T_{UB}s and H_Cs in two specimens of a Shawmere anorthosite sample from site 15.

Site mean C directions are plotted in Figure 9 and averaged in Table 3. The mean direction is D = 64°, I = +2° (k = 18, α_{95} = 13°, N = 6 sites).

Interpretation of the A Magnetization

The A magnetization has generally single-domain like high coercivities. If our interpretation of the soft fraction in Racine tonalites as an unresolved PEF overprint is correct, the A RM also has high unblocking temperatures, mainly >500°C in Shawmere anorthosites and >400°C in Racine tonalites and mafic gneisses. On these grounds, the A is likely to be an older remanence than the B RM, which has low as well as high coercivities and unblocking temperatures.

At the crustal depths where the Racine and especially the Shawmere samples originated, ambient temperatures probably exceeded the Curie temperature of magnetite. For this reason, the A magnetization is unlikely to be primary. Instead we envisage the A component as having been acquired as a thermoremanent magnetization (TRM) during early uplift and cooling of the KSZ, before any crustal tilting occurred (Figure 11a). Maximum depths of burial were estimated by Percival and Card [1983] to be about 15 km for the Racine domain and 23.5 km for the Shawmere anorthosite, although slightly deeper levels are more in keeping with the pressures of about 5 and 8 kbar recorded by metamorphic mineral assemblages on the west and east sides of the KSZ, respectively. We have no reliable information to date about the paleogeothermal gradient during uplift of the KSZ. A model retrograde P-T trajectory by Anovitz and Essene [1988] for the Grenville orogen, which shows similar metamorphic zoning and was uplifted from similar depths, indicates a temperature gradient of ≈30°C/km for depths ≤20 km as uplift was proceeding. Since Grenvillian uplift was slow, it is unlikely that this figure is low because of isotherms being carried upward with the rising crust, as in modern, rapidly uplifted plate margins.

Using 30°C/km as our paleogeothermal gradient, we arrive in Figure 11a at depths of 8 km for the 220°C paleoisotherm and 16.5 km for the 470°C paleoisotherm. Even allowing for the fact that blocking temperatures in nature are lower than those measured in the laboratory for slowly cooled rocks [Pullaiah et al., 1975], these temperatures are certainly low enough to have blocked the A RM as a TRM in the Racine tonalites and gneisses and the Shawmere anorthosite, respectively. About 7 km of uplift would have occurred by the time blocking of the A was complete. Erosion and uplift modeling, constrained by geochronological data from rocks of shallow, intermediate, and deep original crustal levels [Percival et al., 1988], indicates 5-7 km of uplift around 2.6-2.5 Ga, with accompanying cooling to the levels we have assumed above, before final rapid uplift, accompanied by thrusting and crustal tilting, around 2.0-1.9 Ga.

The mean A RM directions from the Racine and Shawmere traverses are distinct at the 95% confidence level (Figure 12(a)), as shown by a Watson [1956] test value of 8.5 when the two traverse means were compared (the significance value is 3.44 [Hartley and Pearson, 1956]). This difference between A directions from the east and west sides of the KSZ might at first sight be taken to indicate a slightly younger age of RM acquisition for Shawmere samples, which were uplifted from a deeper crustal level than the Racine rocks and lagged behind in their cooling. Offsetting this expected trend is the fact

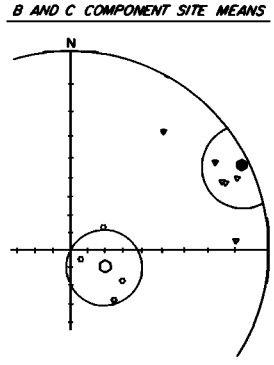


Fig. 9. Site mean B (stars) and C (triangles) directions from the Racine and Shawmere traverses, together with their means and circles of 95% confidence. Open symbols denote upward vectors.

TABLE 2. Component B Site Mean Directions After AF and Thermal Cleaning

Site	N	D, deg	I, deg	k	α_{95}, deg
13	1	139	-53		
15	3	121	-56	36	14
17	5	55	-67	28	12
19	5	131	-82	35	11
Mean	4	115	-68	17	17
Paleopole		144W	49N	dp=24°	dm=28°

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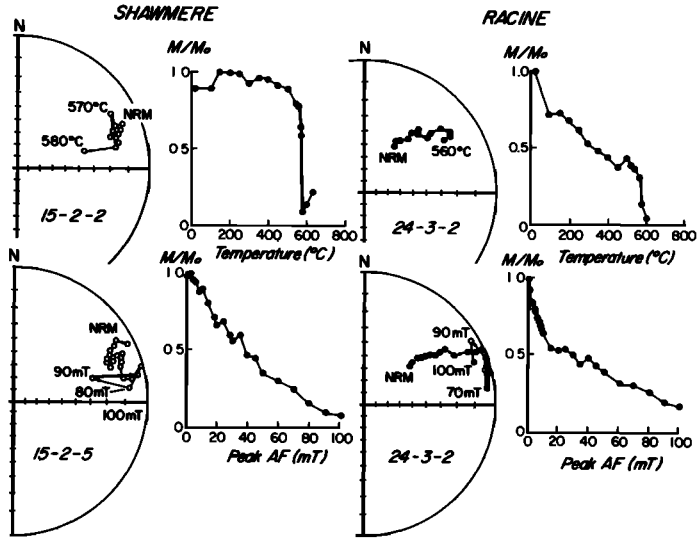


Fig. 10. AF and thermal demagnetization of C RMs in Shawmere anorthosite sample 15-2 and Racine mafic gneiss sample 24-3.

that A blocking temperatures average lower for the Racine rocks than for the Shawmere anorthosite samples. However, when paleomagnetic poles are calculated from the A directions, neither the Racine nor the Shawmere pole lies close to any known part of the Archean or early Proterozoic apparent polar wander path (APWP) for North America (Figure 13).

However, tilt correction of the A directions and their corresponding paleopoles, by amounts that are reasonable in view of the hypothesized structure of the KSZ uplift, do reconcile the Racine and Shawmere results (Figures 12b and 12c), and at the same time place the poles on the APWP around 2.55 Ga (Figure 13). It is the success of these tilt corrections that led us to propose above that the A magnetization was acquired during early (2.6-2.5 Ga) uplift and cooling, but prior to tilting. The fact that the uncorrected Racine and Shawmere paleopoles are displaced with respect to each other in a direction which is mainly perpendicular, not parallel, to the late Archean APWP (Figure 13) suggests that the difference between them has more to do with structure than with a major difference in their ages.

Various combinations of tilt corrections, in 5°

TABLE 3. Component C Site Mean Directions After AF and Thermal Cleaning

Site	N	D, deg	I, deg	k	α_{95}, deg
15	3	66	-10	41	13
19	2	67	-5	17	
35	1	87	+10		
24	1	59	+9		
25	1	38	+16		
32	2	67	-9	16	
Mean	6	64	+2	18	13
Paleopole		154W	18S	dp=7° dm=13°	

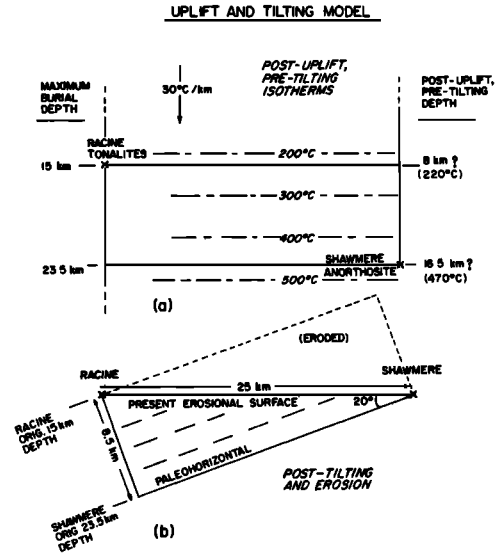


Fig. 11. Model of uplift and crustal tilting in the KSZ. (a) Early slow uplift of ≈7 km, without tilting, around 2.6-2.5 Ga brought the Shawmere anorthosite to the 470°C isotherm and the Racine tonalites to the 220°C isotherm, below the minimum blocking temperatures for the A magnetization in both rocks. The age of this early uplift episode is recorded by the position of structurally corrected A paleopoles on the Laurentian APWP (Figure 13). (b) Geometry of the KSZ after further uplift of 8-16 km, accompanied by 20° crustal tilting, and subsequent erosion. The age of this episode, which probably represents the main Kapuskasing upthrusting event, is not paleomagnetically constrained, but the amount of tilting is recorded by the corrections that must be applied to pretilting Shawmere and Racine A magnetizations to make them consistent with each other (Figure 12) and with the known APWP (Figure 13). The slow rate of early uplift and cooling and the high A unblocking temperatures in Shawmere anorthosite samples have combined to preserve pretilting magnetizations in these deeply exhumed rocks.

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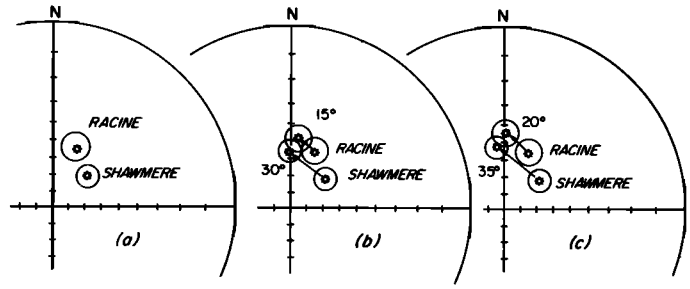


Fig. 12. Comparison of Racine and Shawmere mean A poles (a) before and (b,c) after tilt correction.

increments, for the two traverse means are tabulated in Table 4. Tilting is assumed to have occurred about a fixed horizontal line striking 35°E, parallel to the trend of the KSZ. The minimum Watson test values, corresponding to maximum mutual overlap of confidence circles (Figures 12b and 12c), occur for tilt corrections of 15 — 20° for the Racine area and 30 — 35° for the Shawmere area.

These tilt corrections place the Racine and Shawmere A paleopoles on the 2.6 — 2.5 Ga section of the APWP (Figure 13), near Kenoran-age poles from the Shelley Lake and Burchell Lake granites and the Poohbah Lake alkaline complex of the western Superior Province [Dunlop, 1984a, b, 1985]. The corrected pole positions are significantly displaced from poles of the Matachewan dikes of the Abitibi Subprovince [Irving and Naldrett, 1977; Schutts and Dunlop, 1981; Ernst and Halls, 1984]. The slight difference in corrected positions of the Shawmere and Racine A poles on the APWP may indicate a small difference in mean magnetization ages. However, the difference in pole positions is not significant even at the 90% confidence level.

Is the 15-30° of tilting we have derived paleomagnetically reasonable? Differential tilts for the Racine and Shawmere areas support Percival and Card's [1983] suggestion of a northwesterly flattening of the Kapuskasing structure, although they proposed a somewhat

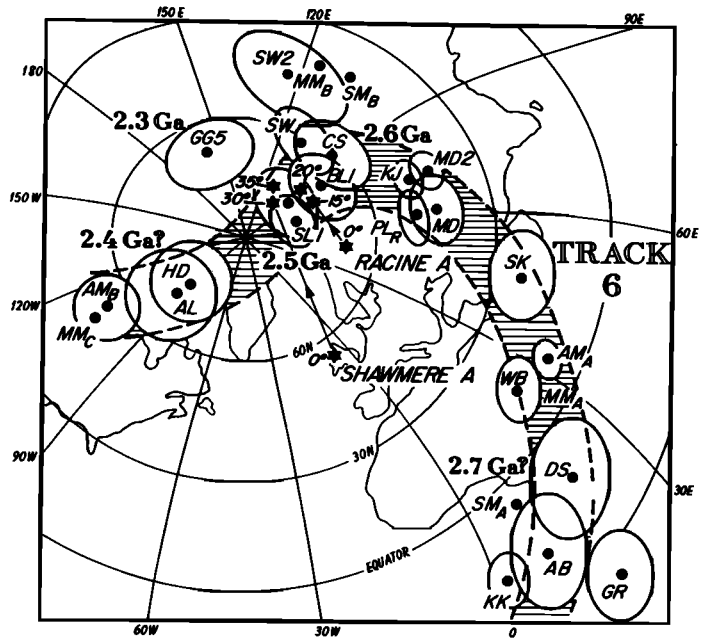


Fig. 13. Late Archean APWP for North America, showing the Racine and Shawmere A paleopoles before and after the tilt corrections of Figure 12. Paleopoles shown, with their ovals of 95% confidence where reported, are KK, DS, MD, Kamiskotia complex, Dundonald sill, and Matachewan diabase, Ontario [Irving and Naldrett, 1977]; GR, Ghost Range intrusive complex, Ontario [Geissman et al., 1982]; AB, MD2, Abitibi basin primary and Matachewan diabase, Ontario [Schutts and Dunlop, 1981]; SM_A, MM_A, AM_A, MM_B, SM_B, AM_B, MM_C, primary and secondary magnetizations of Algoman banded iron formations from the Sherman Mine, Moose Mountain Mine, and Adams Mine, Ontario (Symons and Stupavsky [1983]; also quoted by Séguin et al., [1983]); WB, Wabigoon gabbro, Ontario [Dunlop, 1983]; SK, Skead volcanics, Ontario [Ridler and Foster, 1981]; KJ, Kinojevis metavolcanics, Ontario [Geissman et al., 1983]; PL_R, PL_N (overlain by SL1), reverse and normal polarities, Poohbah Lake alkaline complex, Ontario [Dunlop, 1985]; SW, SW2, Stillwater complex, Montana, after correction for Laramide tilting [Bergh, 1970; Geissman and Mogk, 1986]; CS, Chibougamau sills, Québec [Ueno and Irving, 1976]; SL1, Shelley Lake granite, Ontario, type 1 RM, dated at 2.58±0.02 Ga by $^{40}Ar/^{39}Ar$ on hornblende and biotite [Dunlop, 1984a; Berger and York, 1979]; BL1, Burchell Lake granite, Ontario [Dunlop, 1984b]; HD, AL, Haileybury diorite and Algoman granite, Quebec [Londry and Symons, 1976].

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TABLE 4. Component A Traverse Mean Directions, Watson Test Values and Paleopoles After Various Tilt Corrections

Correction, deg		Racine, deg		Shawmere, deg		Test
Racine	Shawmere	D	I	D	I	value
0	0	23	53	50	62	8.50
25	30	358	42	359	56	5.53
15	20	6	48	13	61	5.47
20	25	1	45	5	58	5.41
10	15	11	50	21	62	5.20
10	20	11	50	13	61	3.63
20	30	1	45	359	56	3.42
30	40	354	39	349	49	3.37
25	35	358	42	353	52	3.33
5	15	17	52	21	62	3.13
15	25	6	48	5	58	3.00
25	40	358	42	349	49	2.65
20	40	1	45	349	49	2.61
20	35	1	45	353	52	2.55
15	30	6	48	359	56	2.47

Correction, deg		Racine Paleopole				Shawmere Paleopole			
Racine	Shawmere	Long-itude	Lat-itude	dp, deg	dm, deg	Long-itude	Lat-itude	dp, deg	dm, deg
0	0	39E	68N	7	10	4W	55N	6	8
20	35	94E	69N	6	9	119E	74N	5	7
15	30	81E	70N	6	9	102E	78N	5	7

smaller amount of crustal tilt. According to preliminary reflection seismic data [Cook, 1985], the ILCZ shallows with depth. For this reason, we take 20° as an average tilt of the Chapleau block across the 20- to 25-km distance separating the Racine and Shawmere traverses. As Figure 11b indicates, an average 20° dip across this distance is perfectly compatible with Percival and Card's [1983] estimate of 8.5 km of contrast in the pretilting depths of these regions.

Beck [1980] has argued from a geometrical picture similar to that of Figure 11b, that "primary" (i.e., preuplift and tilting) magnetizations probably cannot survive block tilting which results in differential uplift of more than about 4 km at either end of the block. His arguments turn on two assumptions: uplift is rapid and carries isotherms upward, and magnetization blocking occurs only below 400°C, whatever the rock type. Beck's arguments are cogent in the setting of microplate or other block rotations at recently active plate margins, where the temperature gradient may be high, burial and subsequent uplift are likely to be rapid, and the lithologies affected are heterogeneous. The burial and uplift of Precambrian orogens seem to have followed an entirely different style, with comparatively slow (≥100 Ma) rebound and moderate, quasi-equilibrium temperatures during uplift from 15- to 25-km depths. This difference in style and the fortunate circumstance that our deepest level rocks, the anorthosites, have discrete high blocking temperatures, has permitted the A RMs to survive much greater postmagnetization tilting and contrast in exposed crustal depths than would be possible in Tertiary orogens.

Finally, we examine the consistency between our estimate of the time of initial uplift and other evidence. Preliminary $^{40}Ar/^{39}Ar$ hornblende ages for tonalitic and mafic gneisses in the same region range from 2.58 to 2.40 Ga [Farrar and Archibald, 1985; Hanes, 1985; Lopez-Martinez and York, 1986]. This range is similar to the 2.6 — 2.5 Ga age span deduced paleomagnetically. The blocking temperature for Ar diffusion in hornblende is 500° — 600°C, while the A component unblocking temperatures were mainly between 500° and 580°C. Thus the agreement between the radiometric and paleomagnetic data is very good.

Interpretation of the B and C Magnetizations

The A component appears to be the oldest surviving RM in our KSZ samples. If so, the B and C components must be younger overprints. There are unfortunately very few examples of bivectorial A+B magnetizations. Figure 14

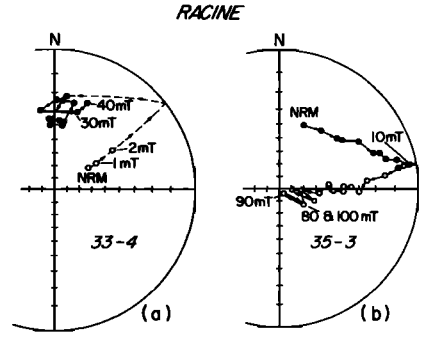


Fig. 14. Examples of superimposed A + B NRMs in tonalitic gneiss samples from sites 33 and 35. A and B are the characteristic NRMs in Figures 14a and 14b, respectively.

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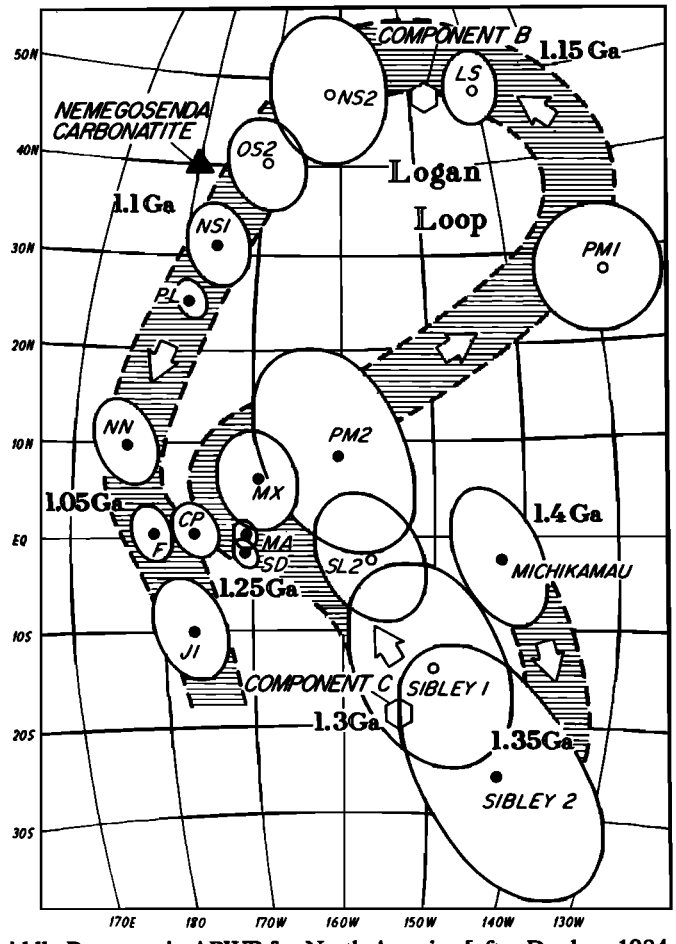


Fig. 15. Part of the middle Proterozoic APWP for North America [after Dunlop, 1984a; Halls and Pesonen, 1982], showing the paleopoles of components B and C of this study and of the Nemegosenda carbonatite [Symons and Garber, 1974].

gives two instances of such behavior for tonalitic gneiss samples from sites 33 and 35. In one case (33-4), A is the ChRM isolated by AF demagnetization and B is a low-coercivity (≤4 mT) overprint. In the other case (35-3), A again has relatively high coercivities (≤40 mT), but B is harder still and is isolated as the ChRM.

If B were a partial TRM thermally overprinting the A magnetization, it would always have lower T_{UB}s, and normally also lower H_Cs, than the A RM. It must be that B is not wholly a TRM but in part a chemical remanent magnetization (CRM). The growth or recrystallization of magnetite could create high laboratory T_{UB}s for the B component which overlap and obscure T_{UB}s of the A component in samples like 35-3.

The C component, which occurs infrequently, poses a different problem. Although apparently stable, it lies in the plane formed by the A and B vectors and is intermediate between them. C could therefore be an unresolved resultant of A and B remanences with very similar H_C and T_{UB} spectra, or a CRM acquired at the same time as B in a direction intermediate between a preexisting A NRM and the B field direction [e.g., Heider and Dunlop, 1987]. For the moment, however, let us assume that C is a distinct RM component.

The B antipole (Table 2) at 144°W, 49°N, and the C antipole (Table 3) at 154°W, 18°S fall on the APWP around 1.1 and 1.3 Ga, respectively (Figure 15). The pole positions date the times of magnetization acquisition. Both RMs postdate crustal tilting.

The B antipole resembles, in both direction and polarity, reverse polarity paleopoles from 1.15 — 1.1 Ga Keweenawan volcanics of the Lake Superior region, immediately west of the KSZ. The B antipole is also similar to the normal polarity paleopole of the Nemegosenda carbonatite [Symons and Garber, 1974], which lies between the Shawmere main body and the Racine traverse (Figure 2) and has been dated by K/Ar on nepheline at 1.0 Ga [Gittins et al., 1967].

The B component could be a thermochemical overprint related to carbonatite intrusion in the KSZ around 1.1 — 1.0 Ga, and in a more general fashion to Keweenawan rifting and volcanism about the same time. In support of this idea, the B component is most prominent at Shawmere anorthosite sites close to the Nemegosenda intrusion. However, the direct heating effect would have been insignificant beyond one intrusion width from the Nemegosenda plug. The B overprinting in all likelihood is due to general warming of, and hydrothermal circulation within, the KSZ carbonatite rift zone. K/Ar and $^{40}Ar/^{39}Ar$ ages of feldspars have been widely updated to 1.2-1.0 Ga in the southern KSZ [Farrar and Archibald, 1985; Lopez-Martinez and York, 1986].

RMs of B affinity in some dikes near the ILCZ were interpreted by Ernst and Halls [1984] as being a secondary overprint in Matachewan dikes but the primary remanence of Kapuskasing dikes. However, a negative baked contact test on one Kapuskasing dike cutting the Shawmere anorthosite [Vandall, 1986] implies that the B RM is not primary in the Kapuskasing dikes either, but records an

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event that magnetically reset both the anorthosite and the dikes. On the other hand, the B-like RM of a KSZ granulite gneiss predates intrusion of lamprophyre dikes that cut it, based on a positive contact test [Vandall, 1986]. Preliminary ages of 1.0 Ga (K/Ar whole rock) for a lamprophyre dike south of the KSZ and 2.05 Ga ($^{40}Ar/^{39}Ar$ on biotite) for a Kapuskasing dike just west of the KSZ bracket the possible age of the B component.

The antipole for component C lies on the Middle Proterozoic APWP for North America between 1.25 Ga poles associated with the Mackenzie igneous episode and pre-Keweenawan poles of the 1.3 Ga Sibley red beds of Lake Superior. Component C is found at sites across the KSZ and has no obvious relation to either uplift and cooling or rifting of the KSZ. The C antipole does resemble a reverse polarity paleopole (SL2 in Figure 15) for the Archean Shelley Lake granite of the Quetico Subprovince, which Dunlop [1984a] interpreted to be a thermochemical overprint associated with incipient Keweenawan rifting. There is little radiometric evidence of a 1.3 — 1.2 Ga event in the KSZ, however.

Conclusions

The oldest event recorded in the paleomagnetism of rocks of the Chapleau block is uplift and cooling of the southern part of the KSZ. This area reached a temperature of 600 — 550°C between 2.6 and 2.5 Ga, based on positions of A paleopoles after correction for tectonic tilt. RM acquisition by granulite-grade rocks near the ILCZ may have lagged 10-30 Ma behind that of amphibolite-grade rocks 20-25 km to the west, but the age difference between paleopoles is not resolved with confidence.

The next event recorded is crustal tilting, which postdated some or all of the uplift. Paleopoles derived from A RMs in the west and the east sides of the Chapleau block must be "untilted" by 15° and 30°, respectively, about the strike of the KSZ to bring them into near coincidence. This tilt correction also restores both poles to the late Archean/early Proterozoic APWP. Crustal tilting must have occurred after ≈2.55 Ga, when the amphibolite and granulite crustal layers had cooled below about 400° and 500°C, respectively. Crustal tilting was greater in the east near the ILCZ than in the west, as proposed by Percival and Card [1983].

A possible event around 1.3 — 1.2 Ga may be recorded by C RMs, which occur sporadically throughout both traverses. However, this component could also be an unresolved composite of A and B RMs, or a stable intermediate-direction CRM.

The final event, at 1.1 — 1.0 Ga, is recorded by B RMs at Shawmere anorthosite sites close to the circa 1.0-Ga Nemegosenda carbonatite. General warming and hydrothermal circulation along the line formed by a series of carbonatite plugs may have accompanied incipient rifting. This rifting in turn may be genetically related to Keweenawan rifting and volcanism immediately to the south and west.

In the units studied, there is no paleomagnetic record of uplift of the KSZ between 1.85 and 1.65 Ga, as proposed by Percival and Sage [1984]. According to our interpretation of the A paleopoles, stabilization of A RMs by initial uplift and cooling to 550-500°C occurred much earlier, between 2.6 and 2.5 Ga. The 1.85 — 1.65 Ga North American paleopoles are well studied and distinctive in their directions, and could hardly have been overlooked. Later post-Penokean events, at around 1.1 Ga and possibly around 1.3 — 1.2 Ga, seem to have overprinted any magnetic record of early to middle Proterozoic activity.

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